

MOFETs

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Derivation of the $i_D - V_{DS}$

The gate and the channel region form a parallel plate capacitor for which the oxide layer serves as a dielectric.

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

$t_{ox} \rightarrow$ Oxide thickness

$\epsilon_{ox} \rightarrow$ permittivity of the SiO_2

$C_{ox} \rightarrow$ Capacitance per unit area.

$$\epsilon_{ox} = \epsilon_0 \times \epsilon_r$$

$$= 8.854 \times 10^{-12} \times 3.9 = 3.45 \times 10^{-11} \text{ F/m.}$$

To find the charge at small section dx

$$Q = C \cdot V \quad I_D = \frac{dQ}{dt}$$

$$Q = C_{ox} V$$

$$Q = C_{ox} [V_{GS} - V_t - V_x]$$

$$dq = C_{ox} \cdot \omega dx$$

$$\frac{dq}{dx} = -C_{ox} \omega [V_{GS} - V_t - V_x]$$

[dq is -ve charge]

$$\text{Velocity } \frac{dx}{dt} = \frac{\text{length}}{\text{time}} = \text{velocity}$$

The voltage V_{DS} produces electric field

$$E(x) = -\frac{dV(x)}{dx}$$

The electric field $E(x)$ causes the electron charge dq to drift toward the drain with a velocity dx/dt

$$\frac{dx}{dt} = -\mu_n E(x)$$

μ_n is the mobility of electrons

Drain current

$$i = \frac{dq}{dt}$$

$$= dq \cdot \frac{dx}{dt}$$

$$i_D = C_{ox} w [V_{GS} - V_t - V_x] \mu_n \frac{dV(x)}{dx}$$

$$i_D dx = \mu C_{ox} w [V_{GS} - V_t - V_x] dV(x)$$

integrating both sides from $x=0$ to $x=L$
 $V(0)=0$ to $V(L)=V_{DS}$

$$\int_0^L i_D dx = \int_0^{V_{DS}} \mu_n C_{ox} w [V_{GS} - V_t - V(x)] dV(x)$$

$$i_D = \left(\mu_n C_{ox} \right) \frac{w}{L} \left[(V_{GS} - V_t) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

① triode region

$$\text{put } V_{DS} = V_{DS} - V_t$$

$$i_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{DS} - V_t)^2$$

— (2) Saturation region.

Note

$k_n = \mu_n C_{ox} \rightarrow$ Process transconductance parameter (A/V^2)

Consider a process technology for which $L_{min} = 0.4 \mu m$, $t_{ox} = 8 nm$, $\mu_n = 450 cm^2/Vs$ and $V_t = 0.7 V$.

a) Find C_{ox} and k_n'

b) For a MOSFET with $W/L = 8 \mu m / 0.8 \mu m$.

Calculate the values of V_{DS} and V_{DSmin} needed to operate the transistor in the saturation region with a dc current $I_D = 100 \mu A$.

c) For the device in (b), find the value of V_{DS} required to cause the device to operate as a 1000Ω resistor for very small V_{DS} .

Solⁿ.

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-11}}{8 \times 10^{-9}} = 4.32 fF/\mu m^2$$

$$k_n' = \mu_n C_{ox} = 450 (cm^2/Vs) \times 4.32 (fF/\mu m^2)$$

$$450 \times 10^8 (\mu m^2/Vs) \times 4.32 \times 10^{-15} (F/\mu m^2)$$

$$k_n' = 194 \mu A/V^2$$

$$k_n' = 194 \times 10^{-6} (F/Vs)$$

b) For saturation

$$I_D = \frac{1}{2} K_n' \frac{W}{L} (V_{GS} - V_t)^2$$

$$100 = \frac{1}{2} \times 194 \times \frac{8}{0.8} (V_{GS} - 0.7)^2$$

$$V_{GS} - 0.7 = 0.32 \text{ V}$$

$$\boxed{\begin{aligned} V_{GS} &= 1.02 \text{ V} \\ V_{DSmin} &= V_{GS} - V_t = 0.32 \text{ V} \end{aligned}}$$

c) For MOSFET in the triode region

$$I_D = K_n' \frac{W}{L} (V_{GS} - V_t) V_{DS}$$

$$g_{DS} = \frac{V_{DS}}{I_D}$$

$$= \frac{1}{K_n' \left(\frac{W}{L} \right) (V_{GS} - V_t)}$$

$$1000 = \frac{1}{194 \times 10^{-6} \times 10 (V_{GS} - 0.7)}$$

$$1000 = \frac{1}{194 \times 10^{-6} \times 10 (V_{GS} - 0.7)}$$

$$V_{GS} - 0.7 = 0.52 \text{ V}$$

$$V_{GS} = 1.22 \text{ V}$$

$$\boxed{V_{GS} = 1.22 \text{ V}}$$

Finite Output Resistance in Saturation.

w.k.t in triode region

$$i_D = k_n' \frac{W}{L} \left[(V_{GS} - V_t) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

If V_{DS} is sufficiently small

$$i_D = k_n' \frac{W}{L} (V_{GS} - V_t) V_{DS}$$

$$g_{DS} = \frac{V_{DS}}{i_D} = \frac{1}{k_n' \frac{W}{L} (V_{GS} - V_t)}$$

$$g_{DS} = \left[k_n' \frac{W}{L} (V_{GS} - V_t) \right]^{-1}$$

In saturation i_D is independent of V_{DS}

$$i_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$$

$$L \rightarrow L - \Delta L \quad [\text{Channel length modulation}]$$

$$= \frac{1}{2} k_n' \left(\frac{W}{L} \right) (V_{GS} - V_t)^2$$

$$= \frac{1}{2} k_n' \left(\frac{W}{L - \Delta L} \right) (V_{GS} - V_t)^2$$

$$= \frac{1}{2} k_n' \frac{W}{L} \frac{1}{1 - \left(\frac{\Delta L}{L} \right)} (V_{GS} - V_t)^2$$

$$\frac{\Delta L}{L} \ll 1$$

$$= \frac{1}{2} k_n' \frac{W}{L} \left(1 + \frac{\Delta L}{L}\right) (V_{GS} - V_t)^2$$

$\therefore \Delta L = \lambda' V_{GS}$
 λ' is a process technology parameter with the dimensions of $\mu m/V$.

$$i_D = \frac{1}{2} k_n' \frac{W}{L} \left(1 + \frac{\lambda' V_{GS}}{L}\right) (V_{GS} - V_t)^2$$

$$\frac{\lambda'}{L} = \lambda$$

$$\therefore i_D = \frac{1}{2} k_n' \frac{W}{L} \left(1 + \frac{\lambda V_{GS}}{L}\right) (V_{GS} - V_t)^2$$

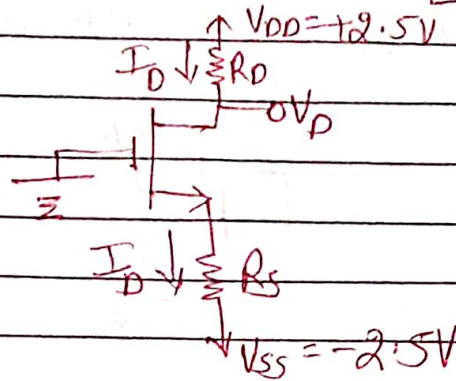
MOSFET Circuit at DC

Neglect the channel length modulation effect
 i.e. $\lambda = 0$

$$V_{GS} - V_t = V_{OV} \text{ [Over drive voltage]}$$

Note!- For NMOS $\rightarrow V_t$ and V_{OV} are positive
 For PMOS $\rightarrow V_t$ and V_{OV} are negative

- 1) Design the circuit of Fig so that the transistor operates at $I_D = 0.4 \text{ mA}$ and $V_D = +0.5 \text{ V}$. The NMOS transistor has $V_t = 0.7 \text{ V}$, $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, $L = 1 \mu\text{m}$, and $w = 32 \mu\text{m}$. [Assume $\lambda = 0$]



Solⁿ: $V_D = 0.5 \text{ V} > V_{th}$ NMOS is in saturation

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{w}{L} (V_{gs} - V_t)^2$$

$$V_{gs} - V_t = V_{ov}, \quad I_D = 0.4 \text{ mA} = 400 \mu\text{A}, \quad \mu_n C_{ox} = 100 \mu\text{A/V}^2$$

$$\frac{w}{L} = \frac{32}{1}$$

$$\therefore 400 = \frac{1}{2} \cdot 100 \times \frac{32}{1} V_{ov}^2$$

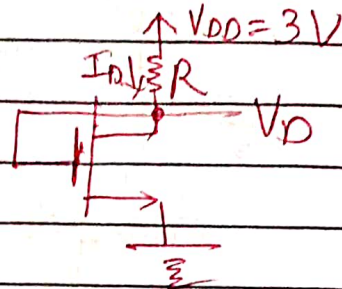
$$V_{ov} = 0.5 \text{ V}$$

$$\therefore V_{gs} = V_t + V_{ov} = 1.2 \text{ V}$$

$$R_S = \frac{V_S - V_{SS}}{I_D} = \frac{-1.2 - (-2.5)}{0.4} = 3.25 \text{ k}\Omega$$

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{2.5 - 0.5}{0.4} = 5 \text{ k}\Omega$$

2) Design the circuit in Fig. 4.21 to obtain a current I_D of $80 \mu A$. Find the value required for R and find the dc voltage V_D . Let the NMOS transistor have $V_t = 0.6V$, $\mu_n C_{ox} = 200 \mu A/V^2$, $L = 0.8 \mu m$ and $w = 4 \mu m$. [Let $\lambda = 0$]



Solⁿ.

$V_{gs} = 0$, $V_D = V_{gs}$ FET is in saturation

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{w}{L} (V_{gs} - V_t)^2$$

$$\therefore I_D = \frac{1}{2} \mu_n C_{ox} \frac{w}{L} (V_{DD})^2$$

$$V_{ov} = \sqrt{\frac{2I_D}{\mu_n C_{ox} (w/L)}}$$

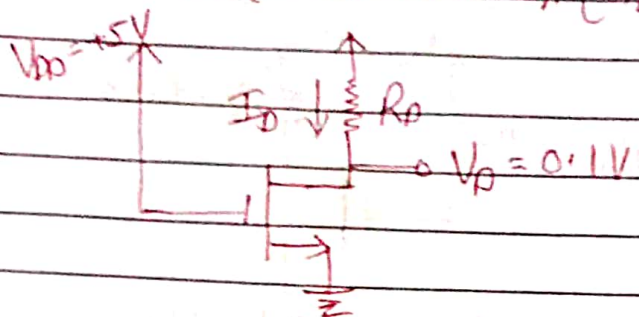
$$V_{ov} = \sqrt{\frac{2 \times 80}{200 (4/0.8)}} = 0.4V$$

$$V_{gs} = V_t + V_{ov} = 0.6 + 0.4 = 1V$$

$$V_D = V_{gs} = 1V$$

$$R = \frac{V_{DD} - V_D}{I_D} = \frac{3 - 1}{0.080} = 25K\Omega$$

- 3) Design the circuit in fig. to establish a drain voltage of $0.1V$. What is the effective resistance between drain and source at this operating point? Let $V_t = 1V$ and $k_n' (W/L) = 1 \text{ mA/V}^2$



Since $V_D < V_G$ and $V_t = 1V$, the MOSFET is in triode region.

$$I_D = k_n' \frac{W}{L} \left[(V_{GS} - V_t) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

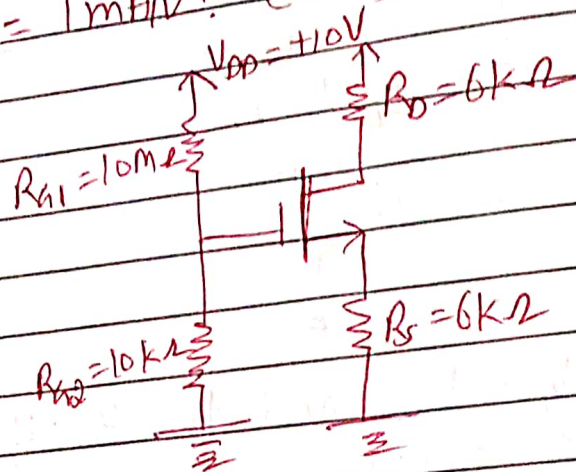
$$I_D = 1 \times \left[(5 - 1) 0.1 - \frac{1}{2} 0.01 \right]$$

$$I_D = 0.395 \text{ mA}$$

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{5 - 0.1}{0.395} = 12.4 \text{ k}\Omega$$

$$r_{DS} = \frac{V_{DS}}{I_D} = \frac{0.1}{0.395} = 253 \Omega$$

4) Analyze the circuit shown in fig. to determine the voltages at all nodes and the currents through all branches. Let $V_t = 1V$ and $k_n'(\frac{W}{L}) = 1mA/V^2$ ($\lambda = 0$)



$$V_G = V_{DD} \cdot \frac{R_{G2}}{R_{G1} + R_{G2}} = \frac{10 \times 10}{10 + 10} = +5V$$

$V_D > V_G$
saturation

$$V_{GS} = 5 - 6I_D$$

$$\begin{cases} V_G = 5V \\ V_S = 6I_D \end{cases}$$

$$I_D = \frac{1}{2} k_n' \left(\frac{W}{L} \right) (V_{GS} - V_t)^2$$

$$= \frac{1}{2} \times 1 \times (5 - 6I_D - 1)^2$$

$$18I_D^2 - 25I_D + 8 = 0$$

$$I_D = 0.89mA \quad \& \quad I_D = 0.5mA$$

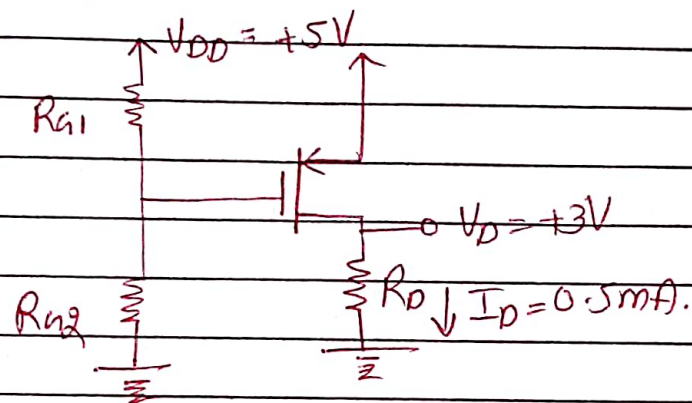
$$I_D \cdot 6 \times I_D \Rightarrow 6 \times 0.89 = 5.34 > V_G$$

i.e. NMOS is in cutoff.

$$I_D = 0.5mA$$

$$V_S = 6I_D = 3V, \quad V_{GS} = 2V, \quad V_D = 10 - 6I_D = 7V$$

- 5) Design the circuit of fig. 50 that the transistor in saturation with $I_D = 0.5 \text{ mA}$ and $V_D = 3 \text{ V}$. Let the enhancement-type PMOS transistor have $V_t = -1 \text{ V}$ and $k_p'(W/L) = 1 \text{ mA/V}^2$. Assume $\lambda = 0$. What is the largest value that R_D can have while maintaining saturation region operation?



solⁿ.
$$I_D = \frac{1}{2} k_p' \frac{W}{L} (V_{GS} - V_t)^2$$

$$= \frac{1}{2} k_p' \frac{W}{L} V_{OV}^2$$

$I_D = 0.5 \text{ mA}$, $k_p'(W/L) = 1 \text{ mA/V}^2$ & PMOS V_{OV} is -ve

$$V_{OV} = -1 \text{ V}$$

$$V_{GS} = V_t + V_{OV} = -1 - 1 = -2 \text{ V}$$

$$R_D = \frac{V_D}{I_D} = \frac{3}{0.5} = 6 \text{ k}\Omega$$

$$V_{D_{\max}} = 3 + 1 = 4 \text{ V}$$

$$R_D = \frac{4}{0.5} = 8 \text{ k}\Omega$$

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